Example 2-6: Measuring Z_L

A ***slotted-line*** probe is an instrument used to measure the unknown impedance of a load, Z_L. A coaxial slotted line contains a narrow longitudinal slit in the outer conductor. A small probe inserted in the slit can be used to sample the magnitude of the electric field and, hence, the magnitude $|\widetilde{V}(d)|$ of the voltage on the line (**Fig. 2-16**). By moving the probe along the length of the slotted line, it is possible to measure $|\widetilde{V}|_{max}$ and $|\widetilde{V}|_{min}$ and the distances from the load at which they occur. Use of Eq. (2.73), namely $S = |\widetilde{V}|_{max}/|\widetilde{V}|_{min}$, provides the voltage standing-wave ratio S. Measurements with a $Z = 50\ \Omega$ slotted line terminated in an unknown load impedance determined that $S = 3$. The distance between successive voltage minima was found to be 30 cm, and the first voltage minimum was located at 12 cm from the load. Determine the load impedance Z_L.

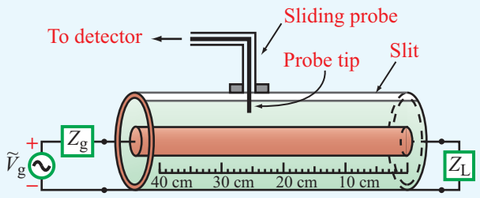


Figure 2-16 Slotted coaxial line (Example 2-6).

Solution: The following quantities are given:

$$Z_0 = 50\ \Omega, \qquad S = 3, \qquad d_{min} = 12\ \text{cm}.$$

Since the distance between successive voltage minima is $\lambda/2$,

$$\lambda = 2 \times 0.3 = 0.6\ \text{m},$$

and

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi}{0.6} = \frac{10\pi}{3} \qquad \text{(rad/m)}.$$

From Eq. (2.73), solving for $|\Gamma|$ in terms of S gives

$$|\Gamma| = \frac{S-1}{S+1} = \frac{3-1}{3+1} = 0.5.$$

Next, we use the condition given by Eq. (2.71) to find θ_r:

$$2\beta d_{min} - \theta_r = \pi, \qquad \text{for } n = 0 \text{ (first minimum)},$$

which gives

$$\theta_r = 2\beta d_{min} - \pi = 2 \times \frac{10\pi}{3} \times 0.12 - \pi = -0.2\pi\ \text{(rad)} = -36^\circ.$$

Hence,

$$\Gamma = |\Gamma| e^{j\theta_r} = 0.5 e^{-j36^\circ} = 0.405 - j0.294.$$

Solving Eq. (2.59) for Z_L, we have

$$Z_L = Z_0 \left[\frac{1+\Gamma}{1-\Gamma}\right] = 50 \left[\frac{1+0.405-j0.294}{1-0.405+j0.294}\right] = (85 - j67)\ \Omega.$$

Exercise 2-12: If $\Gamma = 0.5\angle -60^\circ$ and $\lambda = 24$ cm, find the locations of the voltage maximum and minimum nearest to the load.

Answer: $d_{max} = 10$ cm, $d_{min} = 4$ cm. (See EM.)

Exercise 2-13: A 140 Ω lossless line is terminated in a load impedance $Z_L = (280 + j182)\ \Omega$. If $\lambda = 72$ cm, find (a) the reflection coefficient Γ, (b) the voltage standing-wave ratio S, (c) the locations of voltage maxima, and (d) the locations of voltage minima.

Answer: (a) $\Gamma = 0.5\angle 29^\circ$, (b) $S = 3.0$, (c) $d_{max} = 2.9\ \text{cm} + n\lambda/2$, (d) $d_{min} = 20.9\ \text{cm} + n\lambda/2$, where $n = 0, 1, 2, \ldots$. (See EM.)

2-7 Wave Impedance of the Lossless Line

The standing-wave patterns indicate that on a mismatched line the voltage and current magnitudes are oscillatory with position along the line and in phase opposition with each other. Hence, the voltage to current ratio, called the ***wave impedance*** $Z(d)$, must vary with position also. Using Eqs. (2.63a) and (2.63b) with $z = -d$,

$$Z(d) = \frac{\widetilde{V}(d)}{\widetilde{I}(d)} = \frac{V_0^+[e^{j\beta d} + \Gamma e^{-j\beta d}]}{V_0^+[e^{j\beta d} - \Gamma e^{-j\beta d}]} Z_0 = Z_0 \left[\frac{1+\Gamma e^{-j2\beta d}}{1-\Gamma e^{-j2\beta d}}\right] = Z_0 \left[\frac{1+\Gamma_d}{1-\Gamma_d}\right] \quad (\Omega), \tag{2.74}$$

where we define

$$\Gamma_d = \Gamma e^{-j2\beta d} = |\Gamma| e^{j\theta_r} e^{-j2\beta d} = |\Gamma| e^{j(\theta_r - 2\beta d)} \tag{2.75}$$

as the ***phase-shifted voltage reflection coefficient***, meaning that Γ_d has the same magnitude as Γ, but its phase is shifted by $2\beta d$ relative to that of Γ.

示例 2-6：测量 Z_L

槽线探头是一种用于测量未知负载阻抗 Z_L 的仪器。同轴开槽线在外导体中包含一个狭窄的纵向狭缝。插入狭缝中的小探针可用于对电场强度进行采样，从而对线路上电压的强度 $|\widetilde{V}(d)|$ 进行采样（图 2-16）。通过沿着槽线的长度移动探头，可以测量 $|\widetilde{V}|_{max}$ 和 $|\widetilde{V}|_{min}$ 以及距它们出现的负载的距离。使用等式。(2.73)，即 $S = |\widetilde{V}|_{max}/|\widetilde{V}|_{min}$，提供电压驻波比 S。使用终止于未知负载阻抗的 $Z = 50\ \Omega$ 槽线进行测量，确定 $S = 3$。发现连续电压最小值之间的距离为 30 厘米，第一个电压最小值位于距负载 12 厘米处。确定负载阻抗 Z_L。

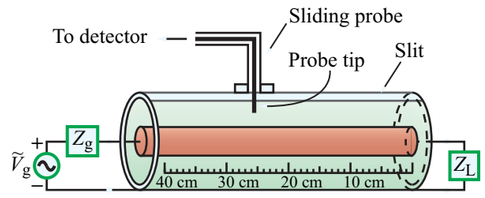


图 2-16 开槽同轴线（示例 2-6）。

解：给出下列数量：

$$Z_0 = 50\ \Omega, \qquad S = 3, \qquad d_{min} = 12\ \text{cm}.$$

由于连续电压最小值之间的距离为 $\lambda/2$，

$$\lambda = 2 \times 0.3 = 0.6\ \text{m},$$

和

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi}{0.6} = \frac{10\pi}{3} \qquad \text{(rad/m)}.$$

从方程。(2.73)，用 S 求解 $|\Gamma|$ 给出

$$|\Gamma| = \frac{S-1}{S+1} = \frac{3-1}{3+1} = 0.5.$$

接下来，我们使用方程给出的条件。(2.71) 求 θ_r：

$$2\beta d_{min} - \theta_r = \pi, \qquad \text{for } n = 0 \text{ (first minimum)},$$

这给出了

$$\theta_r = 2\beta d_{min} - \pi = 2 \times \frac{10\pi}{3} \times 0.12 - \pi = -0.2\pi\ \text{(rad)} = -36^\circ.$$

因此，

$$\Gamma = |\Gamma| e^{j\theta_r} = 0.5 e^{-j36^\circ} = 0.405 - j0.294.$$

求解方程。(2.59) 对于 Z_L，我们有

$$Z_L = Z_0 \left[\frac{1+\Gamma}{1-\Gamma}\right] = 50 \left[\frac{1+0.405-j0.294}{1-0.405+j0.294}\right] = (85 - j67)\ \Omega.$$

练习2-12：如果$\Gamma = 0.5\angle -60^\circ$ 和$\lambda = 24$ cm，找到最接近负载的电压最大值和最小值的位置。

Answer: $d_{max} = 10$ cm, $d_{min} = 4$ cm. (See EM.)

练习 2-13：一条 140 Ω 无损线路的端接负载阻抗为 $Z_L = (280 + j182)\ \Omega$。如果 $\lambda = 72$ cm，求 (a) 反射系数 Γ，(b) 电压驻波比 S，(c) 电压最大值的位置，以及 (d) 电压最小值的位置。

答案：(a) $\Gamma = 0.5\angle 29^\circ$ ✓ (b) $S = 3.0$、(c) $d_{max} = 2.9\ \text{cm} + n\lambda/2$、(d) $d_{min} = 20.9\ \text{cm} + n\lambda/2$，其中 $n = 0, 1, 2, \ldots$。（参见EM。）

2-7 无损线路的波阻抗

驻波模式表明，在不匹配的线路上，电压和电流幅度随着沿线路的位置而振荡，并且彼此反相。因此，电压与电流之比（称为波阻抗 Z(d)）也必须随位置而变化。使用方程式。(2.63a) 和 (2.63b) 与 $z = -d$，

$$Z(d) = \frac{\widetilde{V}(d)}{\widetilde{I}(d)} = \frac{V_0^+[e^{j\beta d} + \Gamma e^{-j\beta d}]}{V_0^+[e^{j\beta d} - \Gamma e^{-j\beta d}]} Z_0 = Z_0 \left[\frac{1+\Gamma e^{-j2\beta d}}{1-\Gamma e^{-j2\beta d}}\right] = Z_0 \left[\frac{1+\Gamma_d}{1-\Gamma_d}\right] \quad (\Omega), \tag{2.74}$$

我们在哪里定义

$$\Gamma_d = \Gamma e^{-j2\beta d} = |\Gamma| e^{j\theta_r} e^{-j2\beta d} = |\Gamma| e^{j(\theta_r - 2\beta d)} \tag{2.75}$$

作为相移电压反射系数，这意味着 Γ_d 与 Γ 具有相同的幅度，但其相位相对于 Γ 移动了 $2\beta d$。

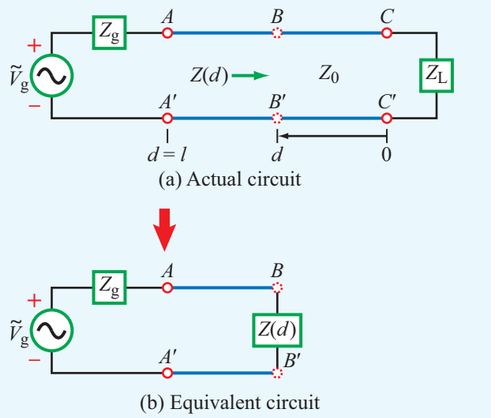


Figure 2-17 The segment to the right of terminals BB' can be replaced with a discrete impedance equal to the wave impedance $Z(d)$.

► $Z(d)$ is the ratio of the total voltage (incident- and reflected-wave voltages) to the total current at any point d on the line, in contrast with the characteristic impedance of the line Z_0, which relates the voltage and current of each of the two waves individually ($Z_0 = V_0^+/I_0^+ = -V_0^-/I_0^-$). ◄

In the circuit of **Fig. 2-17(a)** at terminals BB' at an arbitrary location d on the line, $Z(d)$ is the wave impedance of the line when "looking" to the right, i.e., towards the load. Application of the equivalence principle allows us to replace the segment to the right of terminals BB' with a lumped impedance of value $Z(d)$, as depicted in **Fig. 2-17(b)**. From the standpoint of the input circuit to the left of terminals BB', the two circuit configurations are electrically identical.

Of particular interest in many transmission-line problems is the ***input impedance*** at the source end of the line where $d = l$, which is given by

$$Z_{\text{in}} = Z(d = l) = Z_0 \left[\frac{1+\Gamma_l}{1-\Gamma_l}\right] \tag{2.76}$$

with

$$\Gamma_l = \Gamma e^{-j2\beta l} = |\Gamma| e^{j(\theta_r - 2\beta l)}. \tag{2.77}$$

By replacing Γ with Eq. (2.59) and using the relations

$$e^{j\beta l} = \cos\beta l + j\sin\beta l, \tag{2.78a}$$

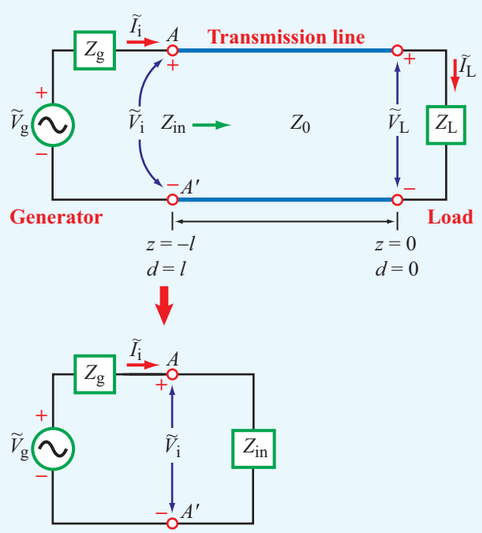


Figure 2-18 At the generator end, the terminated transmission line can be replaced with the input impedance of the line Z_{in}.

$$e^{-j\beta l} = \cos\beta l - j\sin\beta l, \tag{2.78b}$$

Eq. (2.76) can be written in terms of the normalized load impedance z_L as

$$Z_{\text{in}} = Z_0 \left(\frac{z_L \cos\beta l + j\sin\beta l}{\cos\beta l + jz_L \sin\beta l}\right) = Z_0 \left(\frac{z_L + j\tan\beta l}{1 + jz_L \tan\beta l}\right). \tag{2.79}$$

From the standpoint of the generator circuit, the transmission line can be replaced with an impedance Z_{in}, as shown in **Fig. 2-18**. The phasor voltage across Z_{in} is given by

$$\tilde{V}_i = \tilde{I}_i Z_{\text{in}} = \frac{\tilde{V}_g Z_{\text{in}}}{Z_g + Z_{\text{in}}}. \tag{2.80}$$

Simultaneously, from the standpoint of the transmission line, the voltage across it at the input of the line is given by Eq. (2.63a) with $z = -l$:

$$\tilde{V}_i = \tilde{V}(-l) = V_0^+ [e^{j\beta l} + \Gamma e^{-j\beta l}]. \tag{2.81}$$

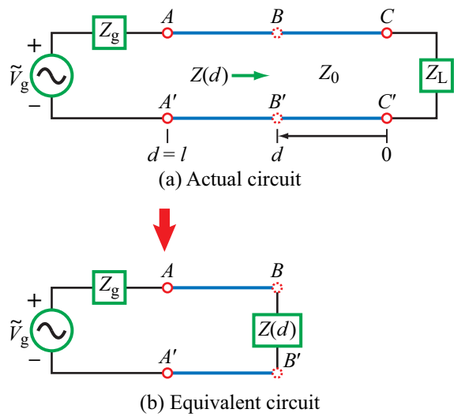


图 2-17 端子 BB′ 右侧的部分可以用等于波阻抗 Z(d) 的离散阻抗代替。

► Z(d) 是线路上任意点 d 处的总电压（入射波电压和反射波电压）与总电流之比，与线路 Z 的特性阻抗相反0，它将两个波的电压和电流分别关联起来（Z 0 =V0+/我0+ =−V0−/我0− ）. ◄

在图 2-17(a) 的电路中，线路上任意位置 d 处的端子 B B′ 处，Z(d) 是向右“看”（即朝负载）时线路的波阻抗。应用等效原理，我们可以将端子 BB′ 右侧的段替换为 Z (d) 值的集总阻抗，如图 2-17(b) 所示。从端子 BB′ 左侧的输入电路的角度来看，两个电路配置在电气上是相同的。

在许多传输线问题中特别令人感兴趣的是线路源端的输入阻抗，其中 d = l，由下式给出

$$Z_{\text{in}} = Z(d = l) = Z_0 \left[\frac{1+\Gamma_l}{1-\Gamma_l}\right] \tag{2.76}$$

和

$$\Gamma_l = \Gamma e^{-j2\beta l} = |\Gamma| e^{j(\theta_r - 2\beta l)}. \tag{2.77}$$

将 Γ 替换为等式。(2.59) 并使用关系式

$$e^{j\beta l} = \cos\beta l + j\sin\beta l, \tag{2.78a}$$

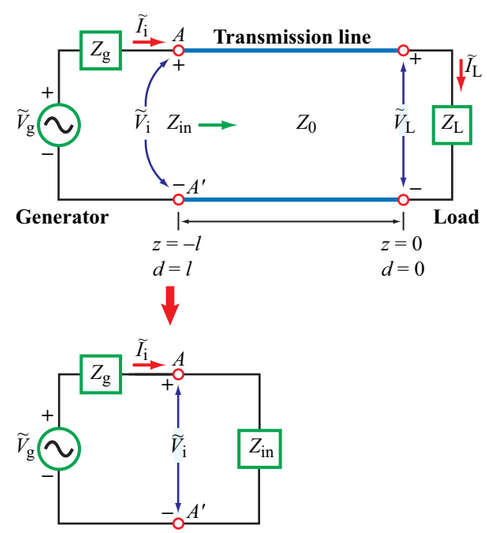


图2-18 在发生器端，端接的传输线可以用线路的输入阻抗Z_{in}代替。

$$e^{-j\beta l} = \cos\beta l - j\sin\beta l, \tag{2.78b}$$

等式。(2.76) 可以用归一化负载阻抗 z_L 写成

$$Z_{\text{in}} = Z_0 \left(\frac{z_L \cos\beta l + j\sin\beta l}{\cos\beta l + jz_L \sin\beta l}\right) = Z_0 \left(\frac{z_L + j\tan\beta l}{1 + jz_L \tan\beta l}\right). \tag{2.79}$$

从发生器电路的角度来看，传输线可以用阻抗Z_{in}代替，如图2-18所示。 Z_{in} 上的相量电压由下式给出

$$\tilde{V}_i = \tilde{I}_i Z_{\text{in}} = \frac{\tilde{V}_g Z_{\text{in}}}{Z_g + Z_{\text{in}}}. \tag{2.80}$$

同时，从传输线的角度来看，传输线输入端的电压由式（1）给出。(2.63a) 与 z = −l：

$$\tilde{V}_i = \tilde{V}(-l) = V_0^+ [e^{j\beta l} + \Gamma e^{-j\beta l}]. \tag{2.81}$$

Equating Eq. (2.80) to Eq. (2.81) and then solving for V_0^+ leads to

$$V_0^+ = \left(\frac{\widetilde{V}_g Z_{in}}{Z_g + Z_{in}}\right)\left(\frac{1}{e^{j\beta l} + \Gamma e^{-j\beta l}}\right). \qquad (2.82)$$

This completes the solution of the transmission-line wave equations given by Eqs. (2.21) and (2.23) for the special case of a lossless transmission line. We started out with the general solutions given by Eq. (2.26), which included four unknown amplitudes, V_0^+, V_0^-, I_0^+, and I_0^-. We then determined that $Z_0 = V_0^+/I_0^+ = -V_0^-/I_0^-$, thereby reducing the unknowns to the two voltage amplitudes only. Upon applying the boundary condition at the load, we were able to relate V_0^- to V_0^+ through Γ, and finally, by applying the boundary condition at the source, we obtained an expression for V_0^+.

Example 2-7: Complete Solution for $v(d,t)$ and $i(d,t)$

A 1.05 GHz generator circuit with series impedance $Z_g = 10\ \Omega$ and voltage source given by

$$v_g(t) = 10\sin(\omega t + 30^\circ) \qquad \text{(V)}$$

is connected to a load $Z_L = (100 + j50)\ \Omega$ through a 50-Ω, 67-cm long lossless transmission line. The phase velocity of the line is $0.7c$, where c is the velocity of light in a vacuum. Find $v(d,t)$ and $i(d,t)$ on the line.

Solution: From the relationship $u_p = \lambda f$, we find the wavelength

$$\lambda = \frac{u_p}{f} = \frac{0.7 \times 3 \times 10^8}{1.05 \times 10^9} = 0.2\ \text{m},$$

and

$$\beta l = \frac{2\pi}{\lambda} l = \frac{2\pi}{0.2} \times 0.67 = 6.7\pi = 0.7\pi = 126^\circ,$$

where we have subtracted multiples of 2π. The voltage reflection coefficient at the load is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{(100 + j50) - 50}{(100 + j50) + 50} = 0.45e^{j26.6^\circ}.$$

With reference to **Fig. 2-18**, the input impedance of the line given by Eq. (2.76) is

$$\begin{aligned} Z_{in} &= Z_0\left(\frac{1+\Gamma_l}{1-\Gamma_l}\right) \\ &= Z_0\left(\frac{1+\Gamma e^{-j2\beta l}}{1-\Gamma e^{-j2\beta l}}\right) \\ &= 50\left(\frac{1+0.45e^{j26.6^\circ}e^{-j252^\circ}}{1-0.45e^{j26.6^\circ}e^{-j252^\circ}}\right) = (21.9 + j17.4)\ \Omega. \end{aligned}$$

Rewriting the expression for the generator voltage with the cosine reference, we have

$$\begin{aligned} v_g(t) &= 10\sin(\omega t + 30^\circ) \\ &= 10\cos(90^\circ - \omega t - 30^\circ) \\ &= 10\cos(\omega t - 60^\circ) \\ &= \Re\mathfrak{e}[10e^{-j60^\circ}e^{j\omega t}] = \Re\mathfrak{e}[\widetilde{V}_g e^{j\omega t}] \qquad \text{(V)}. \end{aligned}$$

Hence, the phasor voltage $\widetilde{V}_g$ is given by

$$\widetilde{V}_g = 10e^{-j60^\circ} = \underline{10/-60^\circ} \qquad \text{(V)}.$$

Application of Eq. (2.82) gives

$$\begin{aligned} V_0^+ &= \left(\frac{\widetilde{V}_g Z_{in}}{Z_g + Z_{in}}\right)\left(\frac{1}{e^{j\beta l} + \Gamma e^{-j\beta l}}\right) \\ &= \left[\frac{10e^{-j60^\circ}(21.9 + j17.4)}{10 + 21.9 + j17.4}\right] \\ &\quad \cdot (e^{j126^\circ} + 0.45e^{j26.6^\circ}e^{-j126^\circ})^{-1} \\ &= 10.2e^{j159^\circ} \qquad \text{(V)}. \end{aligned}$$

Using Eq. (2.63a) with $z = -d$, the phasor voltage on the line is

$$\begin{aligned} \widetilde{V}(d) &= V_0^+(e^{j\beta d} + \Gamma e^{-j\beta d}) \\ &= 10.2e^{j159^\circ}(e^{j\beta d} + 0.45e^{j26.6^\circ}e^{-j\beta d}), \end{aligned}$$

and the corresponding instantaneous voltage $v(d,t)$ is

$$\begin{aligned} v(d,t) &= \Re\mathfrak{e}[\widetilde{V}(d)\, e^{j\omega t}] \\ &= 10.2\cos(\omega t + \beta d + 159^\circ) \\ &\quad + 4.55\cos(\omega t - \beta d + 185.6^\circ) \qquad \text{(V)}. \end{aligned}$$

Similarly, Eq. (2.63b) leads to

$$\begin{aligned} \widetilde{I}(d) &= 0.20e^{j159^\circ}(e^{j\beta d} - 0.45e^{j26.6^\circ}e^{-j\beta d}), \\ i(d,t) &= 0.20\cos(\omega t + \beta d + 159^\circ) \\ &\quad + 0.091\cos(\omega t - \beta d + 185.6^\circ) \qquad \text{(A)}. \end{aligned}$$

等式(2.80) 至方程式(2.81) 然后求解 V_0^+ 导致

$$V_0^+ = \left(\frac{\widetilde{V}_g Z_{in}}{Z_g + Z_{in}}\right)\left(\frac{1}{e^{j\beta l} + \Gamma e^{-j\beta l}}\right). \qquad (2.82)$$

这就完成了方程式给出的传输线波动方程的求解。（2.21）和（2.23）适用于无损传输线的特殊情况。我们从方程式给出的通用解决方案开始。（2.26），其中包括四个未知幅度，V_0^+、V_0^-、I_0^+ 和 I_0^-。然后我们确定 $Z_0 = V_0^+/I_0^+ = -V_0^-/I_0^-$，从而仅将未知数减少到两个电压幅度。在负载处应用边界条件后，我们能够通过 Γ 将 V_0^- 与 V_0^+ 联系起来，最后，通过在源处应用边界条件，我们获得了 V_0^+ 的表达式。

例 2-7：υ(d,t) 和 i(d,t) 的完整解

具有串联阻抗 $Z_g = 10\ \Omega$ 和电压源的 1.05 GHz 发生器电路

$$v_g(t) = 10\sin(\omega t + 30^\circ) \qquad \text{(V)}$$

通过50-Ω、67厘米长的无损传输线连接到负载 $Z_L = (100 + j50)\ \Omega$。线的相速度为 0.7c，其中 c 是真空中的光速。找到直线上的 υ(d,t) 和 i(d,t)。

解：根据 $u_p = \lambda f$ 的关系，我们求出波长

$$\lambda = \frac{u_p}{f} = \frac{0.7 \times 3 \times 10^8}{1.05 \times 10^9} = 0.2\ \text{m},$$

和

$$\beta l = \frac{2\pi}{\lambda} l = \frac{2\pi}{0.2} \times 0.67 = 6.7\pi = 0.7\pi = 126^\circ,$$

我们减去了 2π 的倍数。负载处的电压反射系数为

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{(100 + j50) - 50}{(100 + j50) + 50} = 0.45e^{j26.6^\circ}.$$

参考图2-18，线路的输入阻抗由式(1)给出。(2.76) 是

$$\begin{aligned} Z_{in} &= Z_0\left(\frac{1+\Gamma_l}{1-\Gamma_l}\right) \\ &= Z_0\left(\frac{1+\Gamma e^{-j2\beta l}}{1-\Gamma e^{-j2\beta l}}\right) \\ &= 50\left(\frac{1+0.45e^{j26.6^\circ}e^{-j252^\circ}}{1-0.45e^{j26.6^\circ}e^{-j252^\circ}}\right) = (21.9 + j17.4)\ \Omega. \end{aligned}$$

用余弦参考重写发电机电压的表达式，我们有

$$\begin{aligned} v_g(t) &= 10\sin(\omega t + 30^\circ) \\ &= 10\cos(90^\circ - \omega t - 30^\circ) \\ &= 10\cos(\omega t - 60^\circ) \\ &= \Re\mathfrak{e}[10e^{-j60^\circ}e^{j\omega t}] = \Re\mathfrak{e}[\widetilde{V}_g e^{j\omega t}] \qquad \text{(V)}. \end{aligned}$$

因此，相量电压 $\widetilde{V}_g$ 由下式给出

$$\widetilde{V}_g = 10e^{-j60^\circ} = 10/-60^\circ \qquad \text{(V)}.$$

方程的应用(2.82) 给出

$$\begin{aligned} V_0^+ &= \left(\frac{\widetilde{V}_g Z_{in}}{Z_g + Z_{in}}\right)\left(\frac{1}{e^{j\beta l} + \Gamma e^{-j\beta l}}\right) \\ &= \left[\frac{10e^{-j60^\circ}(21.9 + j17.4)}{10 + 21.9 + j17.4}\right] \\ &\quad \cdot (e^{j126^\circ} + 0.45e^{j26.6^\circ}e^{-j126^\circ})^{-1} \\ &= 10.2e^{j159^\circ} \qquad \text{(V)}. \end{aligned}$$

使用方程式(2.63a) 对于 $z = -d$，线路上的相量电压为

$$\begin{aligned} \widetilde{V}(d) &= V_0^+(e^{j\beta d} + \Gamma e^{-j\beta d}) \\ &= 10.2e^{j159^\circ}(e^{j\beta d} + 0.45e^{j26.6^\circ}e^{-j\beta d}), \end{aligned}$$

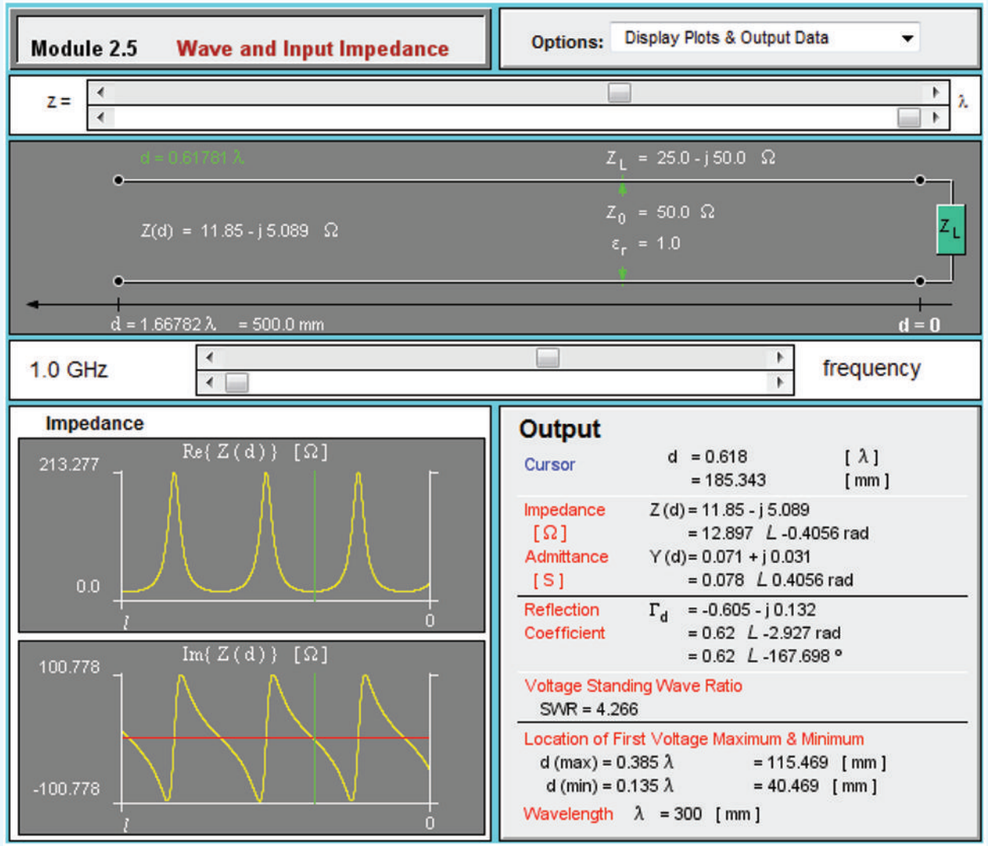
对应的瞬时电压υ(d,t)为

$$\begin{aligned} v(d,t) &= \Re\mathfrak{e}[\widetilde{V}(d)\, e^{j\omega t}] \\ &= 10.2\cos(\omega t + \beta d + 159^\circ) \\ &\quad + 4.55\cos(\omega t - \beta d + 185.6^\circ) \qquad \text{(V)}. \end{aligned}$$

同样，等式。(2.63b) 导致

$$\begin{aligned} \widetilde{I}(d) &= 0.20e^{j159^\circ}(e^{j\beta d} - 0.45e^{j26.6^\circ}e^{-j\beta d}), \\ i(d,t) &= 0.20\cos(\omega t + \beta d + 159^\circ) \\ &\quad + 0.091\cos(\omega t - \beta d + 185.6^\circ) \qquad \text{(A)}. \end{aligned}$$

Module 2.5 Wave and Input Impedance The wave impedance, $Z(d) = \widetilde{V}(d)/\widetilde{I}(d)$, exhibits a cyclical pattern as a function of position along the line. This module displays plots of the real and imaginary parts of $Z(d)$, specifies the locations of the voltage maximum and minimum nearest to the load, and provides other related information.



2-8 Special Cases of the Lossless Line

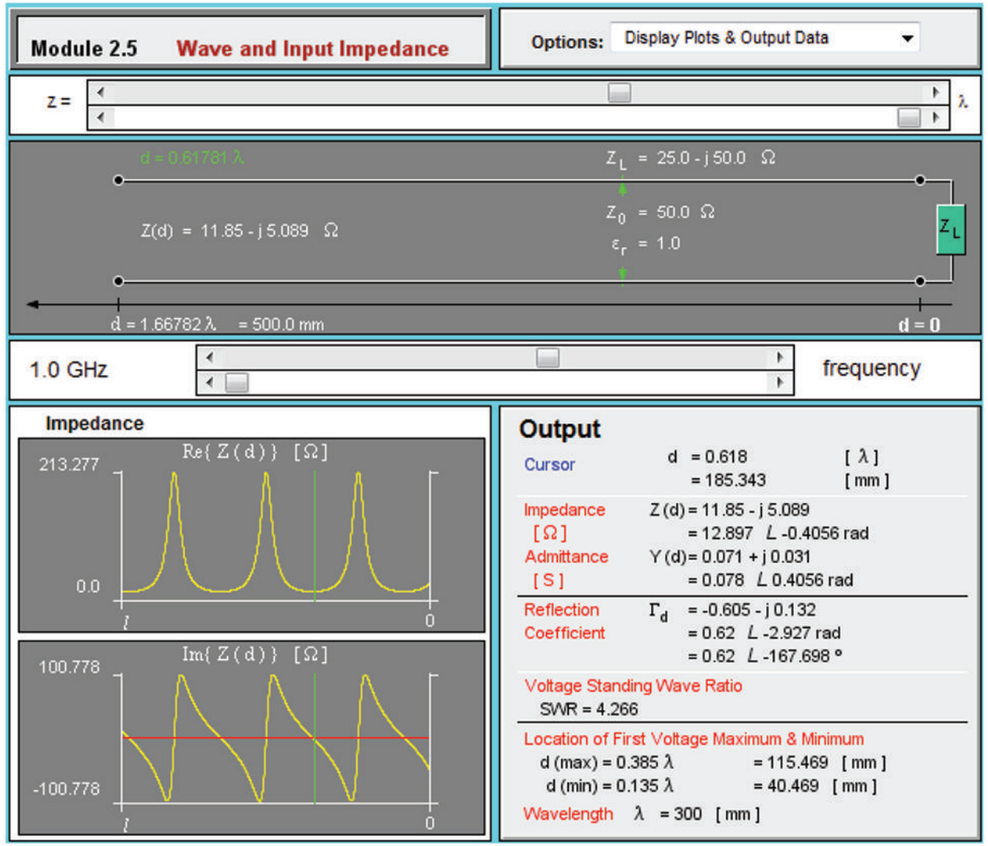
We often encounter situations involving lossless transmission lines with particular terminations or lines whose lengths lead to particularly useful line properties. We now consider some of these special cases.

2-8.1 Short-Circuited Line

The transmission line shown in **Fig. 2-19(a)** is terminated in a short circuit, $Z_L = 0$. Consequently, the voltage reflection coefficient defined by Eq. (2.59) is $\Gamma = -1$, and the voltage standing-wave ratio given by Eq. (2.73) is $S = \infty$. With $z = -d$ and $\Gamma = -1$ in Eqs. (2.63a) and (2.63b), and $\Gamma = -1$ in Eq. (2.74), the voltage, current, and wave impedance on a short-circuited lossless transmission line are given by

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模块 2.5 波和输入阻抗 波阻抗 Z(d) = ˜V (d)/˜I(d) 呈现出作为沿线位置函数的循环模式。该模块显示 Z(d) 的实部和虚部图，指定最接近负载的电压最大值和最小值的位置，并提供其他相关信息。



2-8 无损线路的特例

我们经常遇到涉及具有特定终端的无损传输线或其长度导致特别有用的线路属性的情况。我们现在考虑其中一些特殊情况。

2-8.1 短路线

图 2-19(a) 所示的传输线以短路终止，$Z_L = 0$。因此，电压反射系数由式（1）定义。(2.59)为$\Gamma = -1$，电压驻波比由式(2.59)给出。(2.73) 是 $S = \infty$。式中 $z = -d$ 和 $\Gamma = -1$ （2.63a）和（2.63b），以及等式中的$\Gamma = -1$ （2.74），短路无损传输线上的电压、电流和波阻抗由下式给出

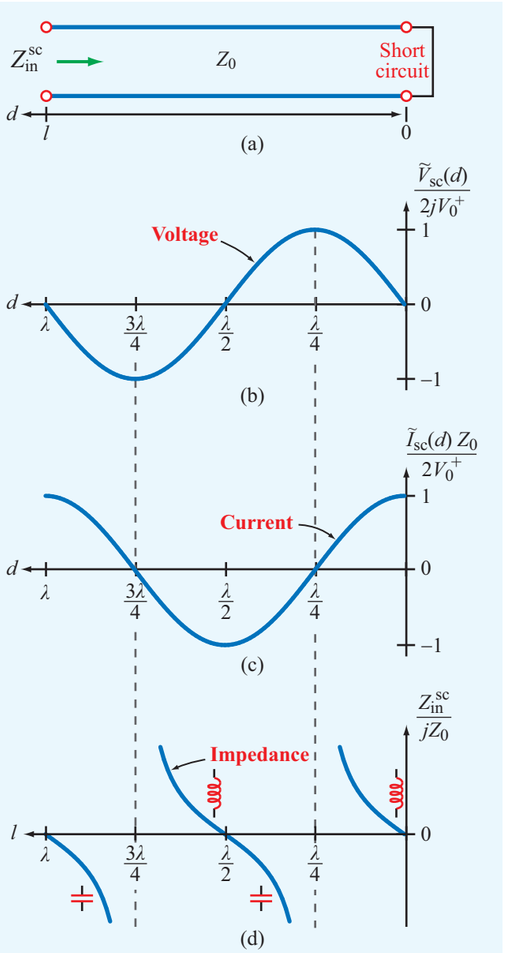


Figure 2-19 Transmission line terminated in a short circuit: (a) schematic representation, (b) normalized voltage on the line, (c) normalized current, and (d) normalized input impedance.

$$\tilde{V}_{sc}(d) = V_0^+[e^{j\beta d} - e^{-j\beta d}] = 2jV_0^+ \sin\beta d, \tag{2.83a}$$

$$\tilde{I}_{sc}(d) = \frac{V_0^+}{Z_0}[e^{j\beta d} + e^{-j\beta d}] = \frac{2V_0^+}{Z_0}\cos\beta d, \tag{2.83b}$$

$$Z_{sc}(d) = \frac{\tilde{V}_{sc}(d)}{\tilde{I}_{sc}(d)} = jZ_0 \tan\beta d. \tag{2.83c}$$

The voltage $\tilde{V}_{sc}(d)$ is zero at the load ($d = 0$), as it should be for a short circuit, and its amplitude varies as $\sin\beta d$. In contrast, the current $\tilde{I}_{sc}(d)$ is a maximum at the load, and it varies as $\cos\beta d$. Both quantities are displayed in **Fig. 2-19** as a function of d.

Denoting Z_{in}^{sc} as the ***input impedance of a short-circuited line*** of length l,

$$Z_{in}^{sc} = \frac{\tilde{V}_{sc}(l)}{\tilde{I}_{sc}(l)} = jZ_0 \tan\beta l. \tag{2.84}$$

A plot of Z_{in}^{sc}/jZ_0 versus l is shown in **Fig. 2-19(d)**. For the short-circuited line, if its length is less than $\lambda/4$, its impedance is equivalent to that of an inductor, and if it is between $\lambda/4$ and $\lambda/2$, it is equivalent to that of a capacitor.

In general, the input impedance Z_{in} of a line terminated in an arbitrary load has a real part, called the ***input resistance*** R_{in}, and an imaginary part, called the ***input reactance*** X_{in}:

$$Z_{in} = R_{in} + jX_{in}. \tag{2.85}$$

In the case of the short-circuited lossless line, the input impedance is purely reactive ($R_{in} = 0$). If $\tan\beta l \geq 0$, the line appears inductive to the source, acting like an equivalent inductor L_{eq} whose impedance equals Z_{in}^{sc}. Thus,

$$j\omega L_{eq} = jZ_0 \tan\beta l, \qquad \text{if } \tan\beta l \geq 0, \tag{2.86}$$

or

$$L_{eq} = \frac{Z_0 \tan\beta l}{\omega} \quad \text{(H)}. \tag{2.87}$$

The minimum line length l that would result in an input impedance Z_{in}^{sc} equivalent to that of an inductor with inductance L_{eq} is

$$l = \frac{1}{\beta}\tan^{-1}\left(\frac{\omega L_{eq}}{Z_0}\right) \quad \text{(m)}. \tag{2.88}$$

Similarly, if $\tan\beta l \leq 0$, the input impedance is capacitive, in which case the line acts like an equivalent capacitor with capacitance C_{eq} such that

$$\frac{1}{j\omega C_{eq}} = jZ_0 \tan\beta l, \qquad \text{if } \tan\beta l \leq 0, \tag{2.89}$$

or

$$C_{eq} = -\frac{1}{Z_0\omega \tan\beta l} \quad \text{(F)}. \tag{2.90}$$

Since l is a positive number, the shortest length l for which $\tan\beta l \leq 0$ corresponds to the range $\pi/2 \leq \beta l \leq \pi$. Hence, the

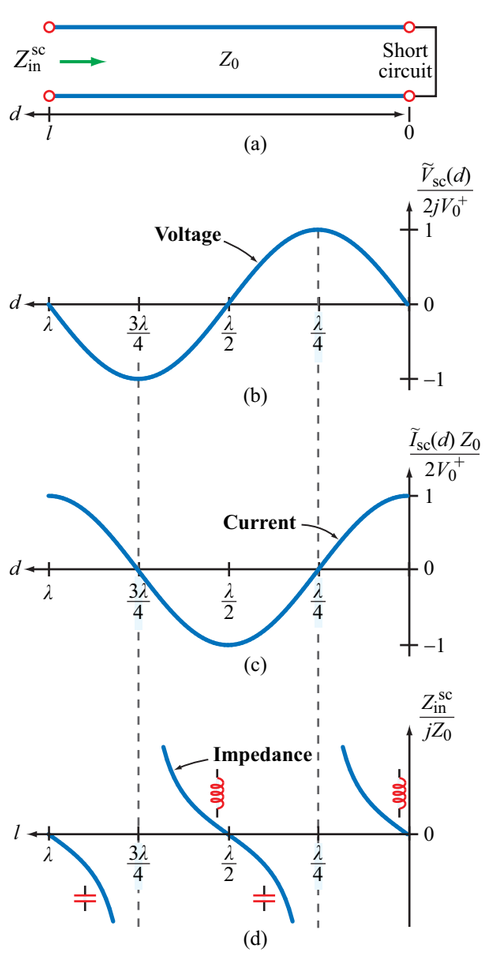


图 2-19 以短路终止的传输线：(a) 示意图，(b) 线路上的标准化电压，(c) 标准化电流，以及 (d) 标准化输入阻抗。

$$\tilde{V}_{sc}(d) = V_0^+[e^{j\beta d} - e^{-j\beta d}] = 2jV_0^+ \sin\beta d, \tag{2.83a}$$

$$\tilde{I}_{sc}(d) = \frac{V_0^+}{Z_0}[e^{j\beta d} + e^{-j\beta d}] = \frac{2V_0^+}{Z_0}\cos\beta d, \tag{2.83b}$$

$$Z_{sc}(d) = \frac{\tilde{V}_{sc}(d)}{\tilde{I}_{sc}(d)} = jZ_0 \tan\beta d. \tag{2.83c}$$

负载处的电压 $\tilde{V}_{sc}(d)$ 为零 ($d = 0$)，短路时应如此，其幅度随 $\sin\beta d$ 变化。相反，电流$\tilde{I}_{sc}(d)$是负载时的最大值，并且随着$\cos\beta d$而变化。这两个量都作为 d 的函数显示在图 2-19 中。

将 Z_{in}^{sc} 表示为长度为 l 的短路线的输入阻抗，

$$Z_{in}^{sc} = \frac{\tilde{V}_{sc}(l)}{\tilde{I}_{sc}(l)} = jZ_0 \tan\beta l. \tag{2.84}$$

Z_{in}^{sc}/jZ_0 与 l 的关系图如图 2-19(d) 所示。对于短路线，如果其长度小于$\lambda/4$，则其阻抗相当于电感，如果在$\lambda/4$和$\lambda/2$之间，则相当于电容器的阻抗。

一般来说，终止于任意负载的线路的输入阻抗 Z_{in} 具有实部（称为输入电阻 R_{in}）和虚部（称为输入电抗 X_{in}）：

$$Z_{in} = R_{in} + jX_{in}. \tag{2.85}$$

在短路无损线路的情况下，输入阻抗是纯电抗性的（$R_{in} = 0$）。如果 $\tan\beta l \geq 0$，则线路对源表现为电感，其作用类似于阻抗等于 Z_{in}^{sc} 的等效电感器 L_{eq}。因此，

$$j\omega L_{eq} = jZ_0 \tan\beta l, \qquad \text{if } \tan\beta l \geq 0, \tag{2.86}$$

or

$$L_{eq} = \frac{Z_0 \tan\beta l}{\omega} \quad \text{(H)}. \tag{2.87}$$

产生与电感为 L_{eq} 的电感器等效的输入阻抗 Z_{in}^{sc} 的最小线路长度 l 为

$$l = \frac{1}{\beta}\tan^{-1}\left(\frac{\omega L_{eq}}{Z_0}\right) \quad \text{(m)}. \tag{2.88}$$

类似地，如果 $\tan\beta l \leq 0$，则输入阻抗为电容性，在这种情况下，线路的作用类似于具有电容 C_{eq} 的等效电容器，使得

$$\frac{1}{j\omega C_{eq}} = jZ_0 \tan\beta l, \qquad \text{if } \tan\beta l \leq 0, \tag{2.89}$$

或者

$$C_{eq} = -\frac{1}{Z_0\omega \tan\beta l} \quad \text{(F)}。 \tag{2.90}$$

由于 l 是正数，因此 $\tan\beta l \leq 0$ 的最短长度 l 对应于范围 $\pi/2 \leq \beta l \leq \pi$。因此，